

UNIVERSITY OF AZAD JAMMU & KASHMIR CHELLAH
CAMPUS MUZAFFERABAD



DEPARTMENT OF SOFTWARE ENGINEERING

Databases Systems

(Lab Task # 04)

Submitted By:

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Submitted To:

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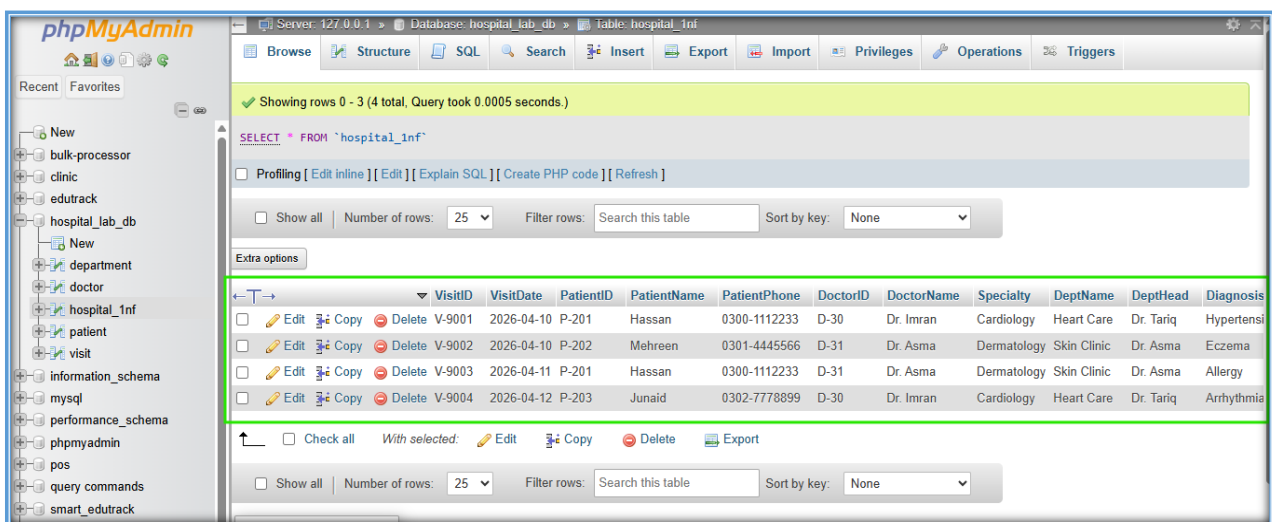
Lab Report: Hospital Management Normalization

➤ Objective:

- To remove data redundancy (repeated information) from the database.
- To organize the data into **3rd Normal Form (3NF)** for better efficiency.
- To ensure that updating one piece of information doesn't require changes in multiple places.

➤ Problem :

- The original table, **hospital_inf**, had all the data mixed into one large sheet.
- **Redundancy:** Patient names and doctor details were repeated every time a visit occurred.
- **Data Risks:** If a doctor changed their specialty, we would have to update every single row manually, which leads to errors.



The screenshot shows the phpMyAdmin interface for a database named 'hospital_lab_db'. The table 'hospital_inf' is selected, and its structure and data are displayed. The table has 12 columns: VisitID, VisitDate, PatientID, PatientName, PatientPhone, DoctorID, DoctorName, Specialty, DeptName, DeptHead, and Diagnosis. The data shows four rows of visits, each with a unique VisitID and VisitDate, but repeating Patient and Doctor information.

VisitID	VisitDate	PatientID	PatientName	PatientPhone	DoctorID	DoctorName	Specialty	DeptName	DeptHead	Diagnosis
V-9001	2026-04-10	P-201	Hassan	0300-1112233	D-30	Dr. Imran	Cardiology	Heart Care	Dr. Tariq	Hypertensi
V-9002	2026-04-10	P-202	Mehreen	0301-4445566	D-31	Dr. Asma	Dermatology	Skin Clinic	Dr. Asma	Eczema
V-9003	2026-04-11	P-201	Hassan	0300-1112233	D-31	Dr. Asma	Dermatology	Skin Clinic	Dr. Asma	Allergy
V-9004	2026-04-12	P-203	Junaid	0302-7778899	D-30	Dr. Imran	Cardiology	Heart Care	Dr. Tariq	Arrhythmia

➤ Normalization:

- **1st Normal Form (1NF):** We ensured every cell contains only one single value (no lists or groups).
- **2nd Normal Form (2NF):** We moved Patient personal details into their own separate table.
- **3rd Normal Form (3NF):** We separated the Department and Doctor info to remove hidden dependencies.

➤ Final Database Structure (The Solution)

We successfully split the messy data into **four organized tables**:

1. **Patient Table:** Stores only patient personal records.
2. **Doctor Table:** Stores names and specialties of the doctors.
3. **Department Table:** Stores hospital departments and their heads.
4. **Visit Table:** The main record that connects everything using IDs (Foreign Keys).

✓ Browse view of the visit table (showing only IDs)

Showing rows 0 - 3 (4 total, Query took 0.0009 seconds.)

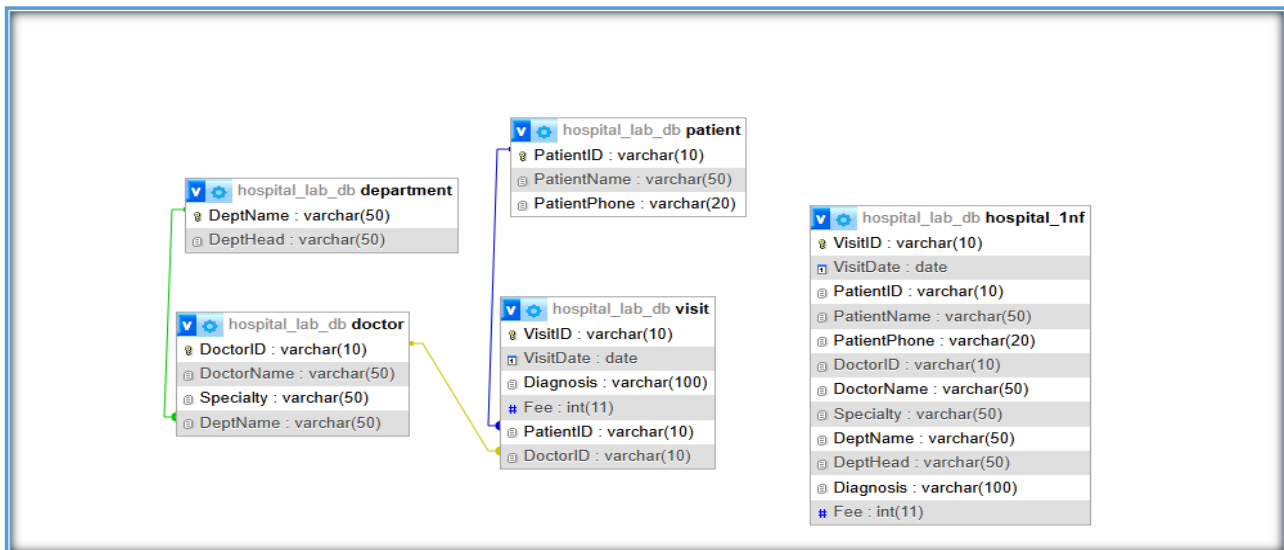
```
SELECT * FROM `visit`
```

Number of rows: 25 Filter rows: Search this table Sort by key: N

	VisitID	VisitDate	Diagnosis	Fee	PatientID	DoctorID
☐ Edit ☐ Copy ☐ Delete	V-9001	2026-04-10	Hypertension	2500	P-201	D-30
☐ Edit ☐ Copy ☐ Delete	V-9002	2026-04-10	Eczema	2000	P-202	D-31
☐ Edit ☐ Copy ☐ Delete	V-9003	2026-04-11	Allergy	2000	P-201	D-31
☐ Edit ☐ Copy ☐ Delete	V-9004	2026-04-12	Arrhythmia	3000	P-203	D-30

Check all With selected: Edit Copy Delete Export

✓ PhpMyAdmin "Designer View" showing the lines connecting the tables



➤ **Implementation Verification:**

To prove the normalization worked, we used an **SQL Join Query** to pull the original data back together instantly without needing to store it redundantly.

Showing rows 0 - 3 (4 total, Query took 0.0009 seconds.)

```
SELECT v.VisitID, v.VisitDate, p.PatientName, d.DoctorName, dept.DeptName, v.Diagnosis, v.Fee FROM visit v JOIN patient p ON v.PatientID = p.PatientID JOIN doctor d ON v.DoctorID = d.DoctorID JOIN department dept ON d.DeptName = dept.DeptName;
```

Number of rows: 25 Filter rows: Search this table

VisitID	VisitDate	PatientName	DoctorName	DeptName	Diagnosis	Fee
V-9001	2026-04-10	Hassan	Dr. Imran	Heart Care	Hypertension	2500
V-9002	2026-04-10	Mehreen	Dr. Asma	Skin Clinic	Eczema	2000
V-9003	2026-04-11	Hassan	Dr. Asma	Skin Clinic	Allergy	2000
V-9004	2026-04-12	Junaid	Dr. Imran	Heart Care	Arrhythmia	3000

Number of rows: 25 Filter rows: Search this table

➤ Conclusion:

- **Successful Normalization:**

The database is successfully transformed from a redundant 1NF structure into a clean, organized 3rd Normal Form (3NF).

- **Elimination of Redundancy:**

By splitting the tables, we ensured that patient and doctor details are no longer repeated unnecessarily, saving storage and reducing errors.

- **Data Integrity:**

Foreign Key constraints were implemented to maintain strong relationships between tables, ensuring that no data is lost during the decomposition.

- **Efficient Updates:**

The new structure allows for "Single-Point Updates." For example, changing a department head now only requires a change in one row rather than multiple visit records.

- **Optimized Querying:**

Using SQL Joins, we proved that complex data can be retrieved instantly without the need for a flat, messy table structure.

- **Scalability:**

This 3NF design makes the hospital system scalable, meaning new doctors, patients, or departments can be added without affecting the existing records.

- **Improved Performance:**

The overall database performance is enhanced as the system handles smaller, specialized tables instead of one massive unorganized dataset.